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% Lab 2 Solutions
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```
% Problem 1
test1 = roots(6, 5, -4);
if (test1(1) == 0)
    type1 = 'real';
else
    type1 = 'complex';
end
fprintf("Solutions to  $6x^2 + 5x - 4 = 0$  are %s and are equal to %.5f and %.5f.\n\n", type1, test1(2), test1(3));

test2 = roots(6*10^154, 5*10^154, -4*10^154);
if (test2(1) == 0)
    type2 = 'real';
else
    type2 = 'complex';
end
fprintf("Solutions to  $(6*10^{154})x^2 + (5*10^{154})x - (4*10^{154}) = 0$  are %s and are equal to %.5f and %.5f.\n\n", type2, test2(2), test2(3));

test3 = roots(0,1,1);
if (test3(1) == 0)
    type3 = 'real';
else
    type3 = 'complex';
end
fprintf("Solutions to  $x + 1 = 0$  are %s and are equal to %.5f and %.5f.\n\n", type3, test3(2), test3(3));

test4 = roots(1,-10^5,1);
if (test4(1) == 0)
    type4 = 'real';
else
    type4 = 'complex';
end
fprintf("Solutions to  $x^2 - (10^5)x + 1 = 0$  are %s and are equal to %.5f and %.5f.\n\n", type4, test4(2), test4(3));

test5 = roots(1, -4, 3.999999);
if (test5(1) == 0)
    type5 = 'real';
else
    type5 = 'complex';
end
fprintf("Solutions to  $x^2 - 4x + 3.999999 = 0$  are %s and are equal to %.5f and %.5f.\n\n", type5, test5(2), test5(3));

test6 = roots(10^(-155), 10^155, 10^155);
if (test6(1) == 0)
    type6 = 'real';
else
    type6 = 'complex';
end
fprintf("Solutions to  $(10^{-155})x^2 + (10^{155})x + (10^{155}) = 0$  are %s and are equal to %.5f and %.5f.\n\n", type6, test6(2), test6(3));

test7 = roots(1, 1, 1);
if (test7(1) == 0)
    type7 = 'real';
else
    type7 = 'complex';
end
fprintf("Solutions to  $x^2 + x + 1 = 0$  are %s and have real and imaginary parts- %.5f and %.5f.\n\n", type7, test7(2), test7(3));

% Problem 2
```

```

x = 1;           % point of evaluation
K = 16;          % determines step sizes
plot_error(x,K);

```

Problem 1

```

% Function to find roots of a quadratic equation

function y = roots(a, b, c)
    % Scaling to avoid overflow and underflow
    s = max([abs(a), abs(b), abs(c)]);

    % Handle the trivial case where all coefficients are zero
    if s == 0
        error('All coefficients are zero; equation is indeterminate.');
```

```

    end

    as = a / s;
    bs = b / s;
    cs = c / s;

    % Handle the Linear Case (a is zero or relatively negligible)
    % Using 'as' here is safer for numerical precision, this also includes
    % underflow.
    if as == 0
        % Equation is bs*x + cs = 0 => x = -cs/bs
        root = -cs / bs;
        y = [0, root, root];
        return;
    end

    % Calculate Discriminant of scaled coefficients
    d_sq = bs^2 - 4*as*cs;

    if d_sq < 0
        % CASE: COMPLEX ROOTS
        type = 1;
        % Real part: -bs / (2*as)
        % Imaginary part: sqrt(-d_sq) / (2*as)
        real_part = -bs / (2 * as);
        imag_part = sqrt(-d_sq) / (2 * as);
        y = [type, real_part, imag_part];
    else
        % CASE: REAL ROOTS
        type = 0;
        % Use the sign of bs to avoid catastrophic cancellation error
        if bs > 0
            Q = -bs-sqrt(d_sq);
            x1 = 2*cs/Q; x2 = Q/(2*as);
        else
            Q = -bs+sqrt(d_sq);
            x1 = Q/(2*as); x2 = 2*cs/Q;
        end
        y = [type, x1, x2];
    end
end

```

Solutions to $6x^2 + 5x - 4 = 0$ are real and are equal to 0.50000 and -1.33333.

Solutions to $(6 \cdot 10^{154})x^2 + (5 \cdot 10^{154})x - (4 \cdot 10^{154}) = 0$ are real and are equal to 0.50000 and -1.33333.

Solutions to $x + 1 = 0$ are real and are equal to -1.00000 and -1.00000.

Solutions to $x^2 - (10^5)x + 1 = 0$ are real and are equal to 99999.99999 and 0.00001.

Solutions to $x^2 - 4x + 3.999999 = 0$ are real and are equal to 2.00100 and 1.99900.

Solutions to $(10^{-155})x^2 + (10^{155})x + (10^{155}) = 0$ are real and are equal to -1.00000 and -Inf.

Solutions to $x^2 + x + 1 = 0$ are complex and have real and imaginary parts- -0.50000 and 0.86603.

Problem 2

```
function y = fin_diff_tan(x,h)
    y = (tan(x+h) - tan(x))/h;
end

function e = error_fd_tan(x,h)
    e = abs(sec(x)^2 - fin_diff_tan(x,h));
end

function y = cent_diff_tan(x,h)
    y = (tan(x+h) - tan(x-h))/(2*h);
end

function e = error_cd_tan(x,h)
    e = abs(sec(x)^2 - cent_diff_tan(x,h));
end

function [] = plot_error(x,K)
    H = 10.^-(0:K);           % step sizes

    error_fd = zeros(size(H));
    error_cd = zeros(size(H));

    for i = 1:length(H)
        error_fd(i) = error_fd_tan(x, H(i));
        error_cd(i) = error_cd_tan(x, H(i));
    end

    [~, fd_idxmin] = min(error_fd);
    [~, cd_idxmin] = min(error_cd);

    % Plot errors
    loglog(H, error_fd, 'r-o', ...
        'LineWidth', 2, 'MarkerSize', 7);
    hold on
    loglog(H(2:7), 1e2*H(2:7).^1, 'k --', 'LineWidth', 2)
    loglog(H, error_cd, 'b-*', ...
        'LineWidth', 2, 'MarkerSize', 7);
    loglog(H(3:6), 1e2*H(3:6).^2 - 10^(-5), 'k -', 'LineWidth', 2)
    loglog(H(fd_idxmin), error_fd(fd_idxmin), 'ko', 'MarkerSize', 10, ...
        'MarkerFaceColor', 'y', 'LineWidth', 2);
    loglog(H(cd_idxmin), error_cd(cd_idxmin), 'ko', 'MarkerSize', 10, ...
        'MarkerFaceColor', 'm', 'LineWidth', 2);

    xlabel('h');
    ylabel('Error');
    title('Finite Difference Error');

    legend( ...
        'Forward difference error', ...
        '$\mathcal{O}(h)$', ...
        'Central difference error', ...
        '$\mathcal{O}(h^2)$', ...
        'Minimum FD error', ...
        'Minimum CD error', ...
        'Location', 'north', ...
        'Interpreter', 'latex' ...
    );
    set(gca, 'FontSize', 10)
    xlim([H(end) H(1)])
    grid on
end
```

